



FASR Science Use Case: Tracking Coronal Cavities for the Pre-eruptive Signatures of Coronal Mass Ejections (Quiet-Sun)

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I. Science Goal(s)

The goal of this science use case is to utilize FASR broadband radio imaging spectroscopy and spectropolarimetry to investigate coronal cavities, which are widely regarded as the coronal manifestations of pre-existing magnetic flux ropes. These observations will diagnose the plasma and magnetic properties of coronal cavities, including their density, temperature structure, and line-of-sight magnetic field, while tracking their temporal evolution from quiescent to pre-eruptive states. A key objective is to determine whether changes in cavity morphology, brightness temperature, radio spectra, polarization, and thermodynamic properties provide measurable precursor signatures of coronal mass ejection (CME) initiation. In cases where cavities erupt, FASR observations will further establish the relationship between pre-eruptive cavity properties and the morphology, dynamics, and radio emission of the resulting CMEs.

II. Scientific Rationale

(A) Scientific Importance

Coronal cavities are dark, elliptical structures often observed above prominences and filament channels. They are widely interpreted as the low-density coronal manifestation of magnetic flux ropes that may later erupt. It is important to emphasize that cavities are not empty but are typically density-depleted relative to the surrounding coronal medium and closely tied to CME initiation and pre-eruptive magnetic topology. Reviews of coronal cavities and prominences are available in [1, 2].

White-light, EUV, and X-ray observations have shown that cavities can reveal important information about their pre-eruptive magnetic configuration [3, 4]. However, the magnetic field within and surrounding the cavity remains difficult to measure using observations at these wavelengths. Recent observations with the Daniel K. Inouye Solar Telescope (DKIST) using circular polarization measurements at IR wavelengths have demonstrated measurable axial magnetic fields on the order of 5-20 Gauss [5]; however, they are limited for several reasons, ranging from small field-of-view to routine monitoring, and mostly for off-limb regions. On the other hand, radio observations provide a unique way to detect cavities both on the disk and at the limb, where they appear as brightness temperature depressions associated with filament channels. [6] showed that metric radio depressions are wider than H-alpha filaments, match EUV filament channels and cavities, and are consistent with density depressions of about 25–50% of the surrounding corona. A recent work by [7] surveyed EUV cavities and provided guidance on the expected occurrence of 3565 coronal cavities over three years, spanning the solar maximum, decline, and minimum phases, yielding average occurrence frequencies of 1188 cavities per year and 3.25 cavities per day.

Recent high-fidelity spectroscopic imaging observations using MeerKAT [8], as shown as an example in Figure 1, provide a new pathfinder example with a coronal cavity observed around 1.28 GHz, exhibiting a clear depression in radio emission with one-to-one correspondence to EUV cavity features, demonstrating the potential of spatially resolved radio imaging spectroscopy to probe cavity plasma properties and motivate similar observations with FASR.

(B) Uniqueness of FASR Capabilities

FASR is uniquely suited for coronal cavity science because, as a dedicated solar observatory, it combines full-Sun, broadband imaging spectroscopy, high-fidelity imaging, and Stokes I and V spectropolarimetry. These capabilities enable simultaneous measurements by monitoring the cavity/cavities over extended periods,

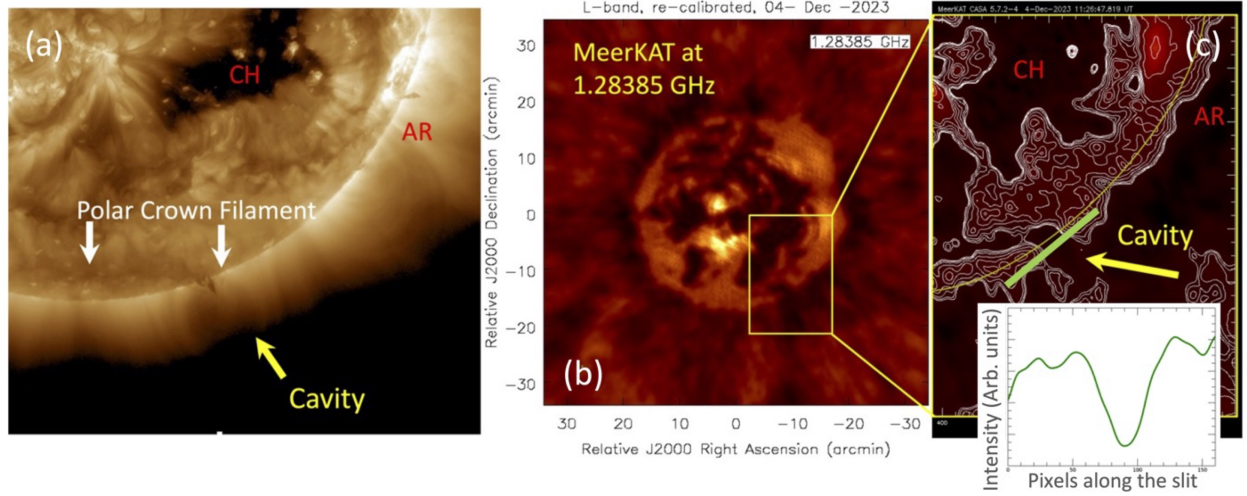


Figure 1: An example of a coronal cavity of a polar crown filament on 2023 December 04. (a) SDO/AIA EUV 193 Å snapshot showing the cavity region at 11:27 UT. (b) Full-Sun observation from MeerKAT array at 1.28 GHz. (c) Zoomed view of the yellow box region in (b) with the contours of 0.3, 1, 2, 6, 10, 20, 40, 60, 80, 90% of the peak emission. The green line denotes the slit used to derive the intensity profile across the cavity. The inset profile demonstrates a distinct reduction in intensity within the cavity relative to the surrounding coronal emission.

continuously tracking their evolution, identifying pre-eruptive signatures, and following their eruption into CMEs.

Past and current radio observations, such as, for example, NRH and EOVSAs, demonstrated the existence and diagnostic potential of radio cavities but were limited by imaging fidelity, spectral sampling, and polarimetric measurements. The radio CME/cavity white paper [9] states that current facilities do not provide adequate imaging fidelity, dynamic range, and polarization purity for routine cavity magnetic diagnostics. Although new generation general-purpose instruments, such as MeerKAT, can provide many of these technical capabilities, they are not dedicated solar observatories and cannot provide the routine, long-term observations required to systematically investigate the evolution of coronal cavities.

FASR can improve this by producing simultaneous multi-frequency imaging spectroscopy and spectropolarimetry, enabling measurements of brightness temperature and circular polarization over a broad range of coronal heights. Stokes V emission of thermal free-free emission can constrain the line-of-sight magnetic field, with expected circular polarization of order a few percent to $\sim 10\%$ depending on observing frequency and magnetic field strength. [9] notes that a 20 G line-of-sight field could produce $\sim 10\%$ circular polarization at 1 GHz. This makes FASR especially valuable for studying both limb cavities and their on-disk counterparts: filament channels, sigmoids, and pre-eruptive flux-rope systems.

(C) Synergies

This SUC has strong synergy with:

- EOVSAs, MeerKAT: pathfinder imaging spectroscopy of cavities.
- SDO/AIA: EUV morphology, DEM temperature estimates, filament channel context.
- MLSO/K-Cor or UCoMP: white-light and coronal magnetic context.

- DKIST/COSMO/UCoMP: optical/IR coronal polarimetry for complementary POS magnetic field constraints.
- LASCO/STEREO/SoloHI/CCOR: tracking whether the observed cavity later erupts as a CME and correlating the CME morphology.
- FORWARD modeling: synthetic radio Stokes I/V emission from MHD flux-rope cavity models.

(D) Measurements Required

For this science objective, FASR should measure spatially resolved brightness temperature and its spectral variation across coronal cavities, filaments, and surrounding streamer regions. These measurements will enable the determination of cavity density depletion, plasma temperature, and thermodynamic structure through thermal free-free emission diagnostics. Stokes I and V observations will provide estimates of the line-of-sight magnetic field with continuous monitoring prior to any potential eruption.

In cases where coronal cavities erupt, FASR observations will also enable the study of the associated CMEs, establishing a direct link between the pre-eruptive cavity and the resulting eruption. Comparing the evolution in various physical parameters of cavities with those of their associated CMEs will help determine how pre-existing magnetic flux-rope structures evolve during eruption and how cavity properties influence CME morphology, dynamics, and radio emission.

III. Science Requirements Tables

The observational requirements listed in Table III are based on the expected properties of coronal cavities and their associated radio emission. Cavities are large-scale structures, typically spanning hundreds of arcseconds, requiring angular resolutions of approximately 10–30 arcsec to resolve cavity boundaries and substructure while maintaining sensitivity to extended emission. Broadband frequency coverage in the range of approximately 0.5–3 GHz is selected to sample thermal free-free emission over multiple coronal heights and to enable imaging spectroscopy and spectropolarimetric diagnostics of the cavity plasma and its magnetic properties. Because cavity emission is expected to vary smoothly with frequency, a spectral resolution of a few MHz is sufficient to characterize brightness temperature spectra and polarization signatures. Temporal resolutions of a few minutes are adequate for tracking cavity evolution, since cavities generally evolve on timescales of hours to days prior to eruption. Polarization accuracy of a few % or better is motivated by the expectation of weak circular polarization signals arising from thermal free-free emission in coronal magnetic fields. The requirement for full-Sun imaging and large spatial scales is driven by the need to monitor multiple cavities simultaneously and to track cavity evolution from quiescent conditions through eruption and the development of associated CMEs.

IV. References

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(A) OBSERVATIONAL TARGET		
Type of observation (what defines a ‘target’?)	Quiet-Sun and pre-eruptive coronal cavities associated with filaments or prominences	
Provide a brief description of the target	Cavities on different limb positions with filaments/polar crown filaments	
Number of targets	Statistical sample (≥ 3 –4 cavities/day) spanning quiescent and pre-eruptive phases	
Size of a single target (arcsec $\times$ arcsec)	a few arcsec	
Distribution of all targets (arcmin $\times$ arcmin)	Full Sun, especially limb and filament-channel regions	
Peak brightness (sfu/beam or Kelvin)	Quite sun level; CME	10^5 – 10^6 K; $\geq 10^6$ K
RMS brightness (sfu/beam or Kelvin)	Sensitivity sufficient to detect cavity depressions of ~ 5 – 10% relative to the surrounding corona	$\lesssim 10^4$ K
Expected circularly polarized flux density	Stokes V (sfu/beam)	-
Expected circular polarization fraction	V/I	~ 0.1 – 10%
Expected linearly polarized flux density	-	
Expected linear polarization fraction	-	

(B) SPECTRAL-TEMPORAL REQUIREMENTS	
Central Frequency (GHz)	~ 0.5 – 3 GHz or above
Instantaneous Bandwidth (GHz/pol)	Broadband frequency coverage
Spectral resolution (MHz)	a few MHz
Temporal resolution (seconds)	$\sim$ a few hours for quiescent tracking, a few seconds for activation/eruption phases

(C) POLARIZATION DATA PRODUCTS REQUIRED	
Stokes I required? (y/n)	y
Stokes Q required? (y/n)	n
Stokes U required? (y/n)	n
Stokes V required? (y/n)	y

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(D) IMAGING REQUIREMENTS		
Required angular resolution (arcsec) (single value or range)	10–30	
Largest angular scale required (arcsec)	a few arcmin	
Mapped image size (arcmin $\times$ arcmin)	a few arcmin	
Required pixel resolution (arcsec)	a few arcsec	
Number of output/image channels	50–500	
Output bandwidth (min–max frequency, GHz)	0.5–3	
Channel width (MHz)	5–20	
Required rms (sfu/beam or Kelvin) [per channel]	Detect $\sim 5\%$ brightness variations	
Dynamic range within image	$\geq 100 : 1$	
Polarization accuracy (%)	few-percent circular polarization (V/I)	Required for line-of-sight magnetic-field measurements
Zero spacing/total power required? (y/n)	n	
Required maximum latency (seconds, or N/A)	N/A	
Required flux density scale calibration accuracy	$\lesssim 10\%$	

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